Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges1

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is **Data Analytics Project: Railway Gauge Trends**
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Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyse historical trends and composition of different railway gauges (Broad, Metre, and Narrow) over the years.

Modules Used:
The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (np.diff).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts)

Project Coverage

Category	Skills Covered
Data Pre-processing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (Easy)

Description:

In this scenario, railway gauge data is imported from a CSV file into a Pandas Data Frame. Missing values are identified and handled to ensure data consistency. Finally, relevant columns are converted into numeric data types to enable accurate analysis and visualization in later stages.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Output:

```
First Five ROWS
      Year  ...  Total
0  1964-65  ...   6876
1  1965-66  ...   6986
2  1966-67  ...   7007
3  1967-68  ...   7027
4  1968-69  ...   7032
```

```
The COLUMN Names
Index(['Year', 'Broad Gauge', 'Metre Gau
ge', 'Narrow Gauge', 'Total'], dtype='st
r')
Year            0
Broad Gauge     0
Metre Gauge     0
Narrow Gauge    0
Total           0
```

```
Data types after conversion:
Year              str
Broad Gauge     int64
Metre Gauge     int64
Narrow Gauge    int64
Total           int64
dtype: object
```

Conclusion:

The dataset was successfully loaded into pandas DataFrame and prepared for analysis, then any missing values found in the data were replaced with 0. The gauge-related columns like Broad, Metre, Narrow, and Total were then converted to number format. The basic data loading and cleaning part were done, now it's ready for further analysis.

Scenario 2: Simple Visualization (Easy)

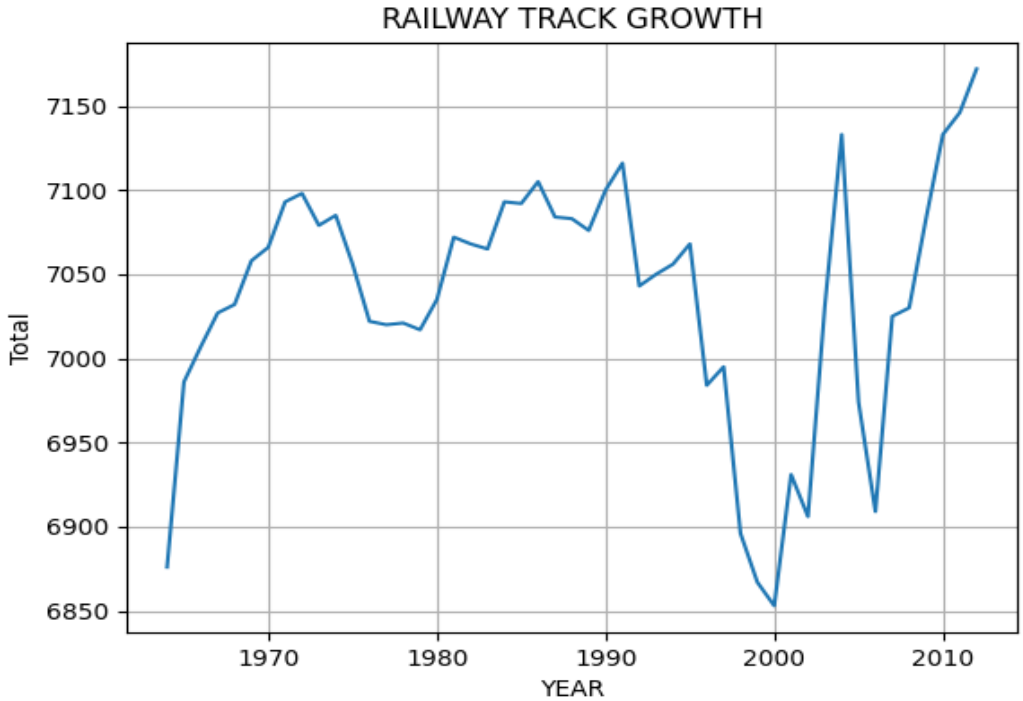
Description:

This scenario focuses on visualizing how India's total railway network has changed in the past years using a line graph. We extract the Year and Total track length columns from the dataset and plot them to see the overall trend. A title and axis labels are added to make the graph understandable.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph:

The graph is a line plot with Year on the X-axis and Total railway track length on the Y-axis. The data points are connected to show how the network changed over time. The track length started low in the 1960, and gradually increased in between the year 1970 and 1980, got down sharply around 2000, and then recovered to all time high after 2010. The chart clearly shows that despite some ups and downs, the overall trend of India's railway network is increasing over the years.



Conclusion:

By observing the graph, the overall trend was identified whether the network expanded or contracted over the years. This simple visualization makes it easy to understand the long-term progress of India's railway infrastructure at just looking at this.

Scenario 3: Filtering + Bar Chart (Medium)

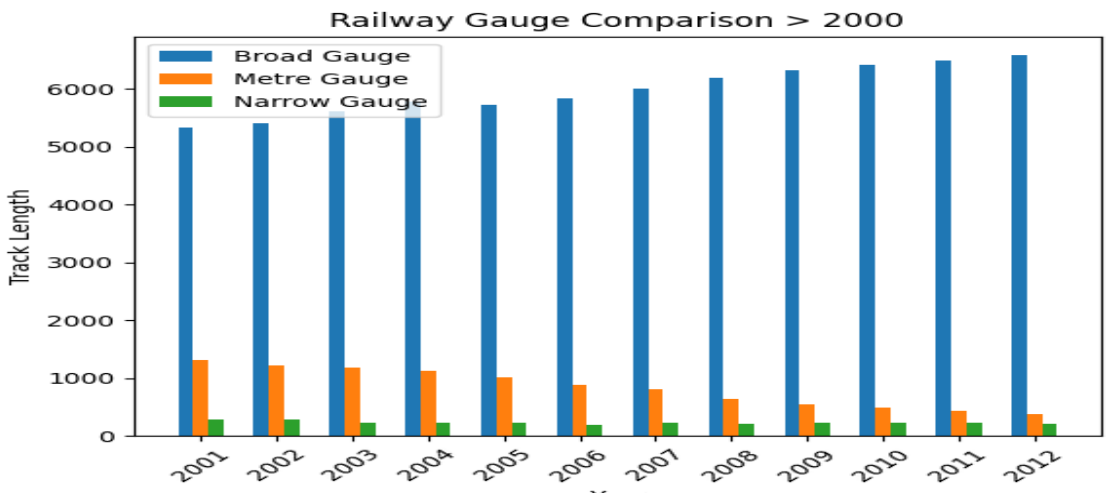
Description:

This scenario focuses on analysing the modern expansion of India's railway network by filtering data for years after 2000. From the filtered data, three gauge (Broad, Metre, and Narrow) and compared using a grouped bar chart

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph:

The graph is a **grouped bar chart** where each year is represented on the X-axis, and the track length is shown on the Y-axis. For every year, three bars are displayed side by side representing Broad Gauge, Metre Gauge, and Narrow Gauge. The chart makes it easy to compare the contribution of each gauge over time.



Conclusion:

After filtering the dataset for years after 2000, it is clear that Broad Gauge dominates the Indian railway network by a very large margin compared to Metre and Narrow Gauge. The grouped bar chart made it easy to visually compare all three gauge types in past years. Metre Gauge and Narrow Gauge showed lower and gradually declining track lengths over the period. This clearly indicates that India has been shifting towards Broad Gauge as the primary standard for its modern railway infrastructure.

Scenario 4: Feature Engineering + Pie Chart (Medium)

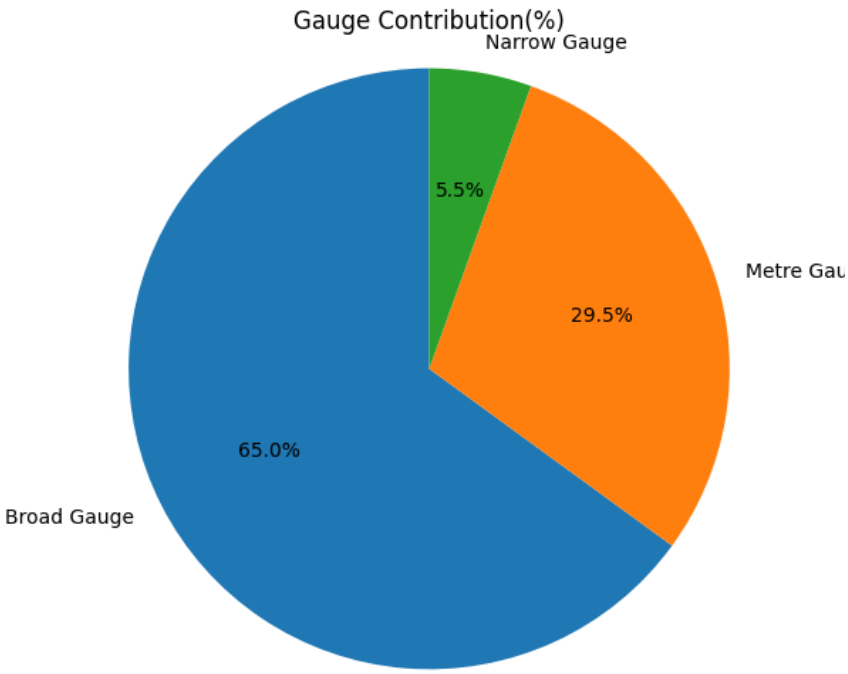
Description:

This scenario focuses on feature engineering to understand the overall contribution of different railway gauges. The total length of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated across all years in the dataset. These aggregated values are stored in a new structure for easy analysis. A pie chart is then used to represent the proportion of each gauge type. This visualization helps in understanding the historical dominance of each gauge in the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:

The graph is a **pie chart** displaying the percentage contribution of Broad Gauge, Metre Gauge, and Narrow Gauge. Each slice represents one gauge type, with percentage labels shown using autopct. The Broad Gauge slice is slightly highlighted (exploded) to emphasize its contribution. The chart provides a clear visual comparison of how much each gauge contributes to the total railway network.



Conclusion:

The pie chart shows that Broad Gauge contributes the more share to the overall railway network. When compared to Metre and Narrow Gauges Broad Gauges is very high in percentage. Metre Gauge holds a smaller portion and Narrow Gauge contributes the least. This indicates that, Broad Gauge has been the most important for the railway system. This analysis show that Broad Gauge is going to be the future.

Scenario 5: Advanced Analysis + Multiple Graphs (Hard)

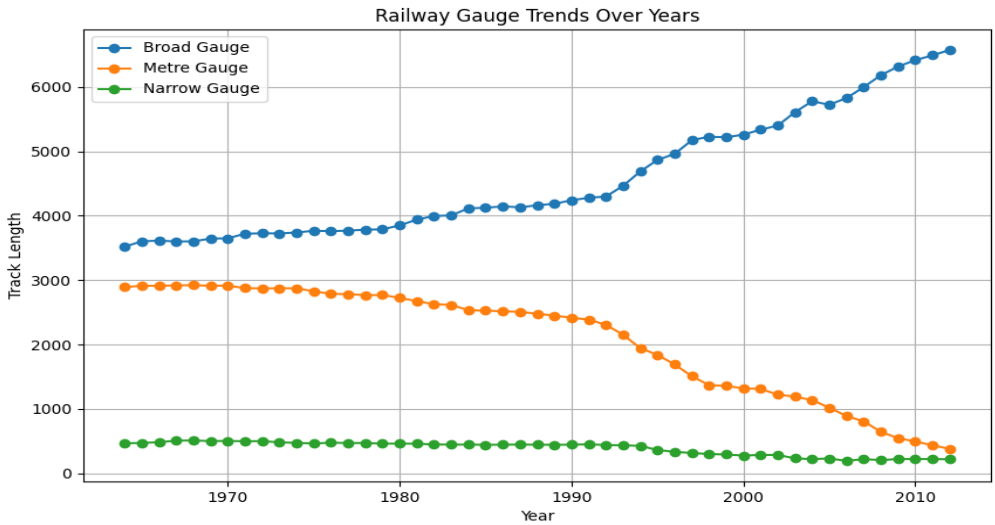
Description:

In this scenario, two types of graphs are used to better understand how the three gauge types have changed during the time. The first is a line graph that shows the percentage share of Broad, Metre, and Narrow Gauge over the years, making it easy to see how each gauge's contribution.

Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graph

The line graph clearly shows how the percentage share of each gauge type has changed over the years. Broad Gauge has been steadily increasing over time, while Metre Gauge and Narrow Gauge have been gradually declining. This indicates that India has been consistently converting and expanding its network towards Broad Gauge. Overall, the line graph gives a clear picture of how the composition of India's railway network has shifted over the decades.



Final Conclusion

This project gave me a clear idea and understanding of how to work with real world data. I have taken the data analysed it and done visualization in many ways. The data clearly shows that Broad Gauge has dominated and growing, while Metre and Narrow Gauge have declined in the years. From all the scenarios, I explored the Datasets like loading, comparing and visualising the graphs, and performing analysis with percentage in pie charts and also with line graphs

What you have learned

By doing this project I learned how to load and clean a real dataset using Pandas and handle problems like missing values and wrong data types. I understood how NumPy helps in doing quick calculations. I also learned how to create different types of graphs like line graphs, bar charts and pie charts using Matplotlib. Most importantly I learned that data analysis is not just about writing code it is about writing the code + making the visuals that are understood by all.